Neck and arm pain assessment

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This tool supports decisions about how soon to seek care and which specialist to see. It does not diagnose or treat, and it does not replace an examination by a doctor. If you feel very unwell or are worried, go to a hospital.

See a spine specialist within a week

Not an emergency, but your symptoms need a proper examination. Go sooner if the pain worsens, numbness spreads or a leg feels weak.

- Arm pain with nerve-root features
- Strong pain (7–8 out of 10)

CLINICAL IMPRESSION

Most likely: Cervical disc herniation or bone spur pinching a nerve root (cervical radiculopathy) [Strong match]

Cervical radiculopathy · disc herniation · foraminal stenosis

Supports: Neck tilt test (gentle Spurling): positive · Middle finger and the middle of the forearm · Symptoms below the elbow · Arm or hand pain is worse · Sharp, electric pain shooting into the arm · Looking up or tilting the head back · Resting the hand on top of the head

A disc or a bony spur in the neck narrows the opening where a nerve root leaves the spine. Pain, tingling and numbness travel down the arm along that root, often worse when looking up or turning the head.

Why your answers point here

- Neck tilt test (gentle Spurling): positive — A positive neck tilt test reproduces root pain by narrowing the opening.
- Middle finger and the middle of the forearm — Middle finger symptoms map to the C7 root, the most commonly affected.
- Symptoms below the elbow — Symptoms below the elbow point to a nerve root rather than a muscle.
- Arm or hand pain is worse — Arm pain worse than neck pain is typical of a pinched nerve root.
- Sharp, electric pain shooting into the arm — Sharp, electric pain running along a line is typical of an irritated nerve root.
- Looking up or tilting the head back — Looking up narrows the openings where neck nerve roots leave the spine.
- Resting the hand on top of the head — Relief with the hand on the head takes tension off a pinched root.

Why it happens: Age-related drying of the discs with bony spurs (spondylosis) causes about two-thirds of cases; a soft disc herniation causes about one in five. Long screen time, heavy overhead work and smoking add risk.

How doctors confirm it: Neurological examination of strength, reflexes and sensation in both arms; MRI of the neck if there is weakness or no improvement after 6 weeks; nerve conduction tests if the cause is unclear.

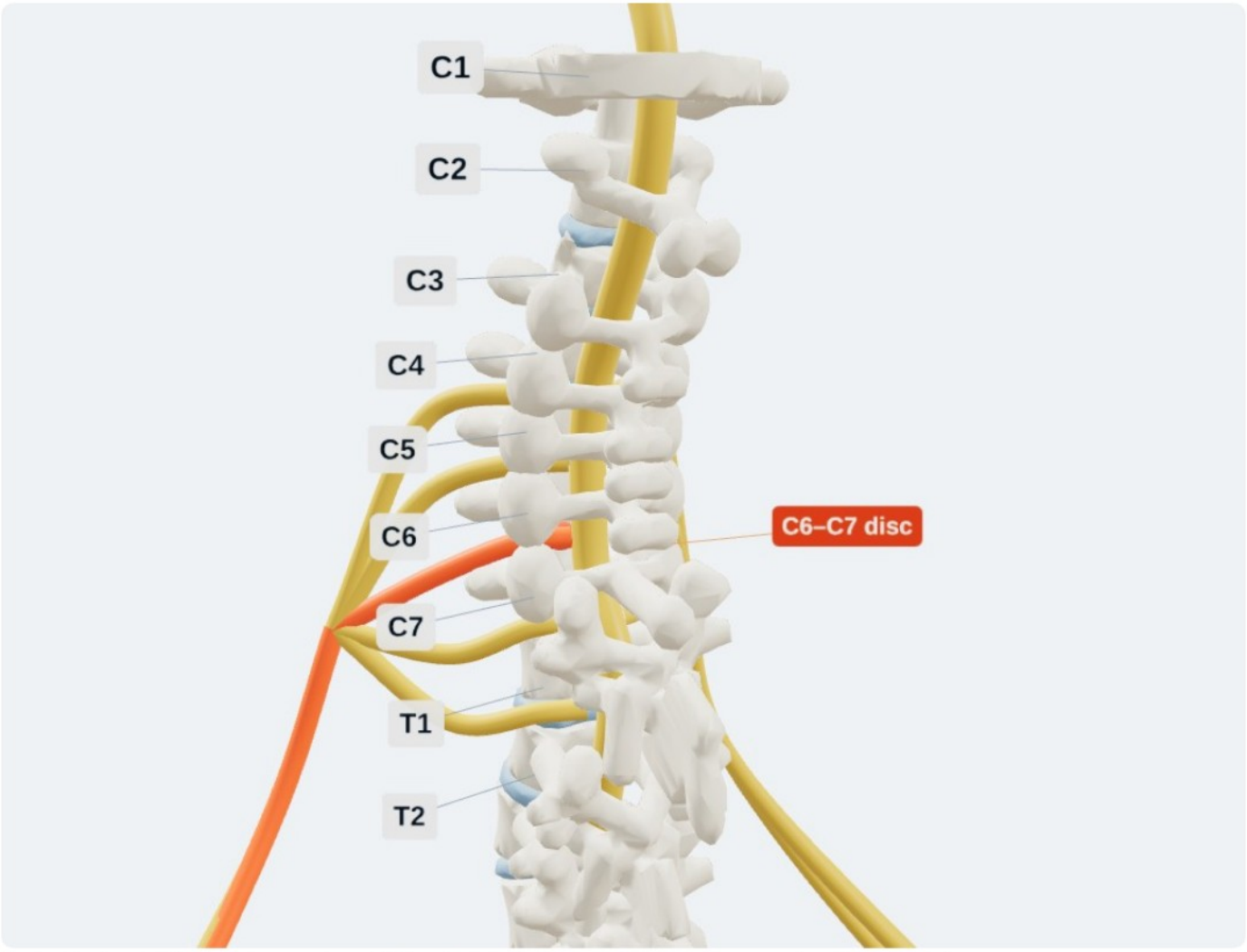
How it is treated: Activity changes, medicines chosen by your doctor, physiotherapy (posture, nerve gliding, strengthening) and sometimes a short period in a soft collar. Image-guided injections for persistent arm pain; surgery (ACDF or disc replacement) for weakness or pain that does not settle.

Outlook: About 90% are symptom-free or only mildly affected at follow-up; about one in four needs surgery, and about one in three has another episode.

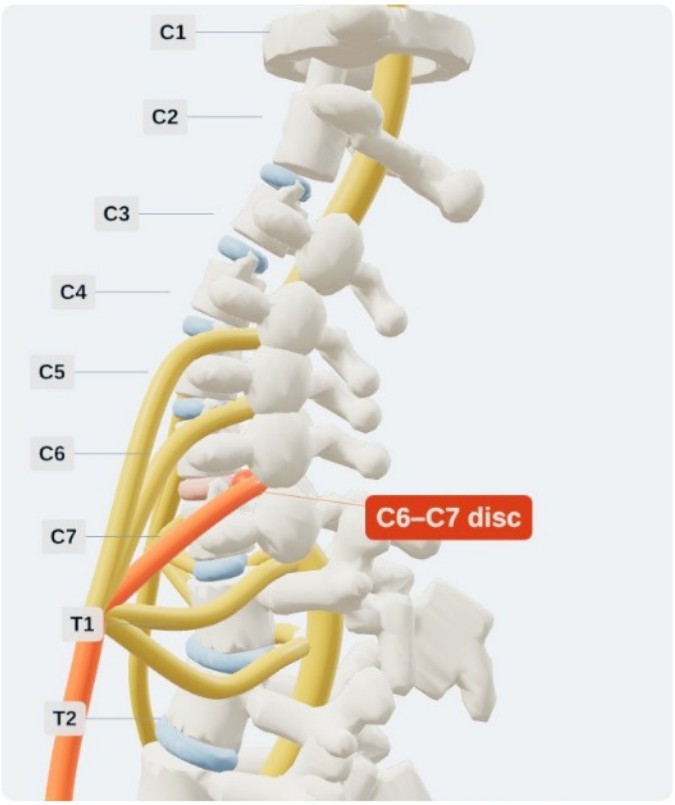
A DOCTOR'S EXAMINATION IS ESSENTIAL

- This assessment cannot test your reflexes, muscle strength or skin sensation, which show which nerve is involved and how badly.
- Only a doctor can decide whether you need an MRI, X-ray or blood tests, and interpret them alongside your examination.
- Similar pain can come from the hip, kidney, pelvis or abdomen; a doctor examines these too.
- Medicines, injections and exercise plans must be chosen for you personally.

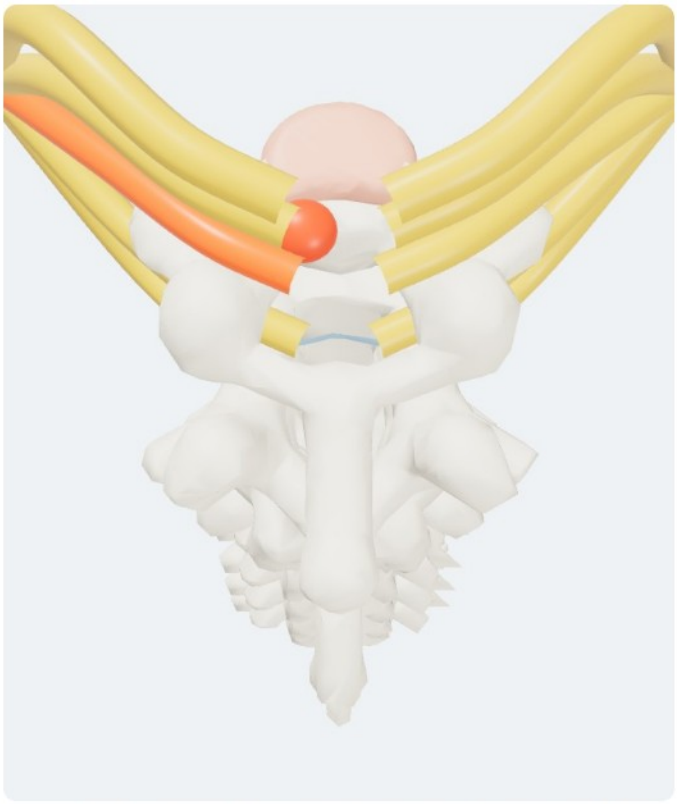
3D ANATOMY OF YOUR PROBLEM



Neck spine from behind, vertebrae numbered. Pink: affected disc. Orange: the pressed nerve root and its path into the arm.



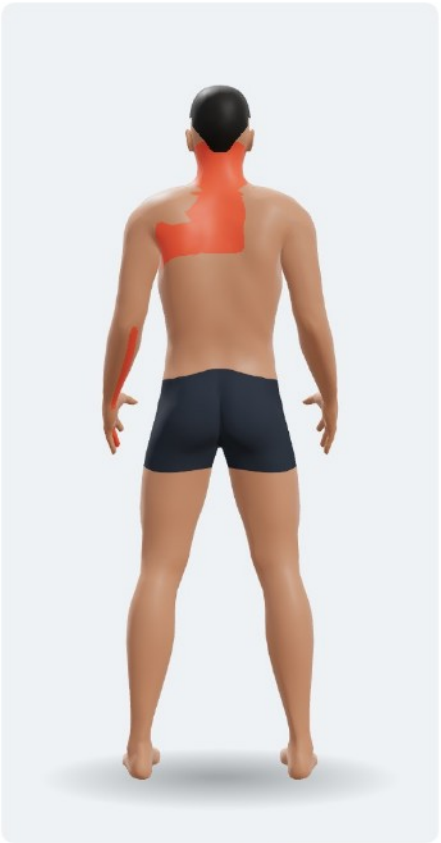
Side view: the disc bulging backwards into the space where the nerve roots run.



Looking down on the C6–C7 disc with the vertebra above removed, as on an MRI cross-section.



Nerve path: the affected root continues down the arm to the hand, which is why symptoms reach the fingers you marked.



Areas you marked, back view



Areas you marked, front view

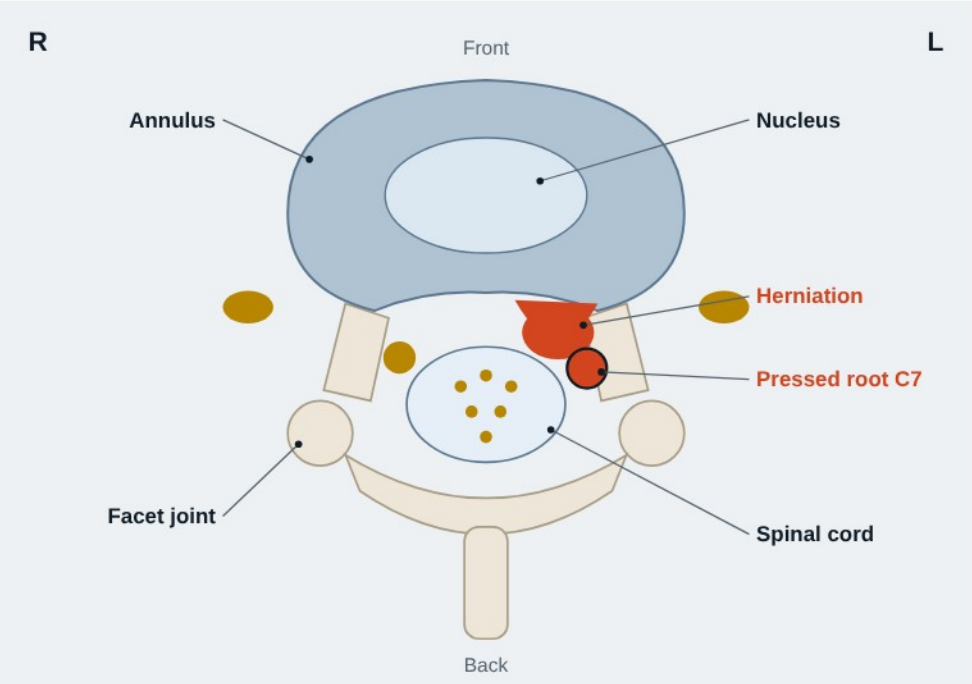
ROOT CAUSE EXPLAINED

Your answers fit a disc or bony spur at C6–C7 pressing on the C7 nerve root on the left side. In the neck, each nerve root leaves the spine through a small opening (foramen) just behind the disc. With age the discs lose water and bony spurs grow (spondylosis), narrowing these openings; a soft disc herniation can also bulge into them.

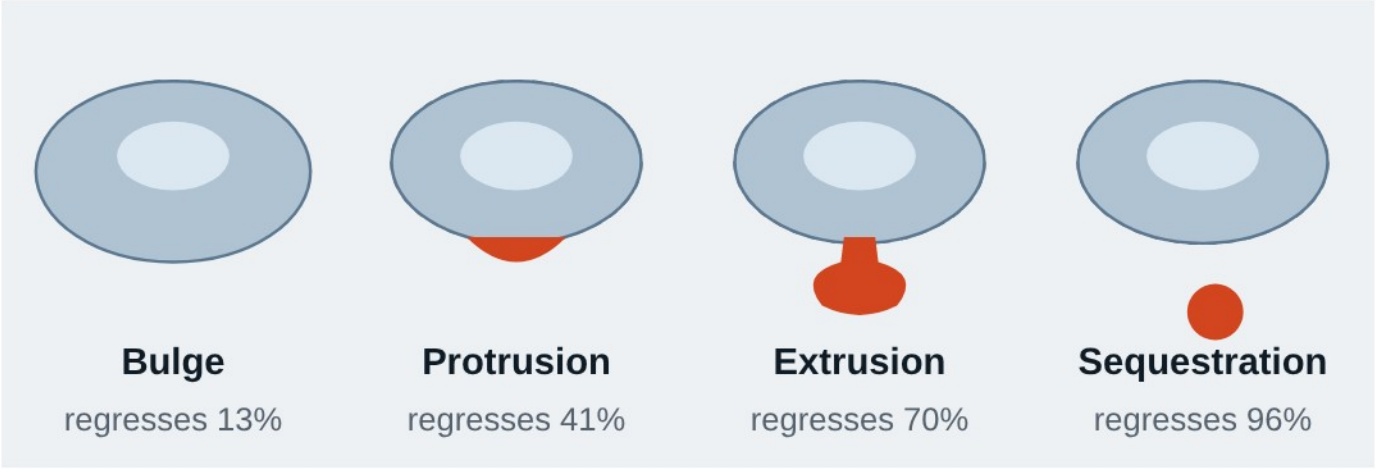
The pressed root becomes inflamed, so pain, tingling and numbness travel down the arm along its own path: C6 to the thumb, C7 to the middle finger, C8 to the little finger. Looking up and tilting the head towards the painful side narrows the opening further, which is why it hurts more; resting the hand on the head takes tension off the root.

Unlike the lower back, the neck canal contains the spinal cord itself. If a disc or spurs press on the cord (myelopathy), the warning signs are clumsy hands, unsteady walking and electric shocks down the spine; these need prompt specialist assessment.

C7 and then C6 are the roots most often affected. About 90% of people are symptom-free or only mildly affected at follow-up, and about one in four needs surgery. Only an MRI confirms the exact level.



Cross-section at the disc, viewed from the feet as on an MRI (your left is on the right).



NERVE ROOT REFERENCE

C7 — Pain and numbness: Middle finger, back of the forearm. Weakness: Straightening the elbow (triceps), wrist flexion. Reflex: Triceps reflex reduced. Usual disc: C6–C7.

WHICH SPECIALIST TO SEE

1. **Orthopaedics – Spine (Start here)**
   Bone, joint and spine specialist; first stop for most back and sciatica problems. For: Cervical disc herniation or bone spur pinching a nerve root (cervical radiculopathy).
2. **Neurosurgery / Spine surgery**
   For nerve compression with weakness, or pain that does not settle with treatment. For: Cervical disc herniation or bone spur pinching a nerve root (cervical radiculopathy).
3. **Neurology**
   Assesses nerve damage; nerve conduction studies and EMG. For: Cervical disc herniation or bone spur pinching a nerve root (cervical radiculopathy).
4. **Physiotherapy / PM&R**
   Structured exercise and rehabilitation; the core of treatment for most back pain. For: Cervical disc herniation or bone spur pinching a nerve root (cervical radiculopathy).

YOUR ANSWERS

Age	40–49 · Male
Medical history (tick any that apply)	None
Areas marked	Back and sides of the neck; Between the shoulder blades; Over the shoulder blade (left); Middle finger and the middle of the forearm (left) [C7]
Which troubles you more: neck pain or arm pain?	Arm or hand pain is worse
How did it start?	Woke up with it
When did it start?	This bout started 1 to 6 weeks ago
How does the pain behave?	Sharp, electric pain shooting into the arm; Pins and needles in the hand or fingers; Looking up or tilting the head back; Resting the hand on top of the head
Impact on daily life	Pain right now 7/10 · Using the arm and hand: With pain or some weakness · Desk, laptop or phone work: Less than 1 hour · Sleep: —
Neck tilt test (gentle Spurling)	Positive
Hand-on-head relief test	Positive
Grip-and-release test (hand speed)	Not answered
Heel-to-toe walk	Not answered
Red-flag screen	No to all warning signs

SOURCES

1. WHO guideline on chronic low back pain (summary, Health Policy Watch, 2023) — https://healthpolicy-watch.news/who-issues-first-guidelines-low-back-pain/
2. Orthobullets – Lumbar disc herniation — https://www.orthobullets.com/spine/2035/lumbar-disc-herniation
3. Chiu CC et al. Probability of spontaneous regression of lumbar herniated disc. Clin Rehabil 2015;29:184–95 (summary) — https://www.painscience.com/biblio/intervertebral-disc-injuries-spontaneously-regress-surprisingly-often--and-worse-ones-more.html

4. Cochrane review CD007431 – Physical examination for lumbar radiculopathy due to disc herniation — https://www.cochrane.org/evidence/CD007431_physical-examination-diagnosis-lumbar-radiculopathy-due-disc-herniation-patients-low-back-pain-and
5. Brinjikji W et al. Imaging features of spinal degeneration in asymptomatic populations. AJNR 2015;36:811–6 — https://www.ajnr.org/content/36/4/811
6. Radiology Assistant – Lumbar disc nomenclature 2.0 — https://radiologyassistant.nl/neuroradiology/spine/lumbar-disc-nomenclature-2-0
7. North American Spine Society – Bulge vs herniation — https://www.spine.org/KnowYourBack/Conditions/Degenerative-Conditions/Bulge-Herniation
8. MSD Manual – Symptoms of common radiculopathies by dermatomal level — https://www.msdmanuals.com/professional/multimedia/table/symptoms-of-common-radiculopathies-by-dermatomal-level
9. Physiopedia – Lumbar radiculopathy assessment (myotomes, reflexes) — https://www.physio-pedia.com/Lumbar_Radiculopathy_Assessment
10. Medmastery – Diagnosing lumbar radiculopathy — https://www.medmastery.com/guide/neurology-diseases-clinical-guide/diagnosing-lumbar-radiculopathy
11. NICE NG59 – Low back pain and sciatica in over 16s — https://www.nice.org.uk/guidance/ng59/chapter/Recommendation
12. Royal Free London NHS – Cauda equina syndrome warning signs — https://royalfree.nhs.uk/patients-and-visitors/patient-information-leaflets/cauda-equina-syndrome
13. WikEM – Back pain red flags — https://wikem.org/wiki/Back_pain_(red_flags)
14. MSD Manual – Lumbar spinal stenosis — https://www.msdmanuals.com/professional/musculoskeletal-and-connective-tissue-disorders/neck-and-back-pain/lumbar-spinal-stenosis
15. ASAS criteria for inflammatory back pain — https://empendium.com/mcmtextbook/table/031_6932
16. Spine-health – Vertebral fracture symptoms — https://spine-health.com/conditions/osteoporosis/vertebral-fracture-symptoms
17. Pott's disease (spinal tuberculosis) — https://en.wikipedia.org/wiki/Pott%27s_disease
18. NHS 111 Wales – Kidney stones — https://111.wales.nhs.uk/kidneystones/
19. Texas Orthopedics – Coccydynia — https://www.txortho.com/patient-education-article/coccydynia-tailbone-pain-969
20. Medical emergency numbers in India (108, 112) — https://zhl.org.in/blog/medical-emergency-numbers-india/
21. GBD 2021 Neck Pain Collaborators. Global burden of neck pain 1990–2020 and projections to 2050. Lancet Rheumatol 2024;6:e142–55 — https://research.uaeu.ac.ae/en/publications/global-regional-and-national-burden-of-neck-pain-19902020-and-pro/
22. Radhakrishnan K et al. Epidemiology of cervical radiculopathy, Rochester 1976–1990. Brain 1994 (PubMed abstract) — https://pubmed.ncbi.nlm.nih.gov/8186959/
23. WikEM – Cervical radiculopathy (exam by level) — https://wikem.org/wiki/Cervical_Radiculopathy
24. Physiopedia – Clinical prediction rule for cervical radiculopathy (Wainner 2003) — https://physio-pedia.com/CPR_for_Cervical_Radiculopathy
25. Orthobullets – Cervical myelopathy — https://www.orthobullets.com/spine/2031/cervical-myelopathy
26. UK Health Security Agency – Meningitis: don’t ignore the signs — https://gov.uk/government/publications/meningitis-signs-and-symptoms-poster/meningitis-dont-ignore-the-signs-and-symptoms
27. Heart attack warning signs (American Heart Association) — https://oaklawnhospital.org/services/cardiology/warning-signs-of-a-heart-attack-american-heart-association/
28. Healthdirect (Australian Government) – Upper back (thoracic) pain — https://www.healthdirect.gov.au/upper-back-thoracic-pain
29. Patient.info – Shingles (herpes zoster) — https://patient.info/health/Shingles.htm
30. Wolfe F et al. 2016 revisions to the fibromyalgia diagnostic criteria (criteria form) — https://lin-web.clarkson.edu/~lrussek/2016FMS.pdf